

# SUSTAINALYZE: A Machine Learning Application For Carbon Emission Forecasting Using Sustainable Energy Indicators

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**Abstract** — This study focuses on predicting carbon dioxide (CO<sub>2</sub>) emissions using a machine learning-based approach aligned with Sustainable Development Goal (SDG) 13: Climate Action. A comprehensive dataset containing socioeconomic, energy, and environmental indicators was utilized to train and evaluate multiple regression models, including Linear Regression, Support Vector Regression, Random Forest, and Gradient Boosting. The models were assessed based on Mean Squared Error (MSE) and R<sup>2</sup> scores to determine predictive accuracy. Initial testing showed that Gradient Boosting achieved the most consistent and reliable performance, with a Test MSE of approximately  $2.16 \times 10^6$  and an R<sup>2</sup> value of 0.9924, outperforming the other models. Further improvement using optimized hyperparameters (Random Search Run 2) resulted in a Test MSE of  $4.43 \times 10^6$  and an R<sup>2</sup> of 0.9844, confirming its robustness across unseen data despite slight variations. The final model was deployed to classify predicted CO<sub>2</sub> emission levels into Low, Medium, or High categories, enabling an interpretable and accessible output for end-users. This predictive system can support policymakers, organizations, and researchers by offering early insights into emission trends based on modifiable energy and development factors.

**Keywords**— Machine Learning, CO<sub>2</sub> Emissions Prediction, Gradient Boosting Regression, Climate Action, Sustainable Development Goal 13, Hyperparameter Optimization, Random Search, Environmental Data Science, Energy Indicators, Regression Modeling

## I. INTRODUCTION

Climate change is one of the most serious problems that the world faces in the 21st century. It is a direct consequence of the continuous rising of carbon dioxide (CO<sub>2</sub>) emissions caused by the energy production and consumption processes. Although there are global efforts for the rapid increase of renewable energy use, the majority of the world's regions are still using fossil fuels extensively; consequently, emission levels remain unsustainable, and the environmental impacts are exacerbated. Worldwide energy and development indicators reveal that differences in access to electricity, energy efficiency, and the use of renewable energy significantly contribute to variations in emissions between countries and levels of the economy [2]. These disparities pinpoint that there is a need for efficient analytical and predictive tools which are capable of understanding the intricate relationships between energy systems and carbon emissions.

Conventional statistical methods usually are not able to detect the nonlinear interactions between various energy, environmental, and socioeconomic factors. It has been proven in several studies that machine learning models, especially ensemble methods, are very efficient in

predicting carbon emissions and finding the variables that have the most significant influence. Deeper studies experimented with algorithms like Random Forest, Support Vector Regressor (SVR), Gradient Boosting Regressors, and Linear Regression on the countries with the highest carbon emissions worldwide. These revealed that factors such as energy consumption, fossil fuel dependency, and GDP were among the most influential contributors to CO<sub>2</sub> emission levels [1]. This evidence advocates for the use of data-driven models as they have the potential to reveal the patterns which cannot be seen by traditional methods of analysis.

As a way of providing easy-to-understand and practical sustainability analytics, this research offers Sustainalyze, a machine learning-powered business analytics application that aims at classifying and predicting carbon emission intensity in different countries and years. Sustainalyze is a feature-rich platform that leverages features such as renewable energy share, electricity access, fossil fuel usage, GDP per capita, etc., to assign a country-year record to a Low, Medium, or High carbon emission category. The system is equipped with a well-orchestrated pipeline for the different stages of the data lifecycle, including preprocessing, model training, function implementation, and visualization, hence providing users with the ability to interactively investigate emission patterns. The application is broadly intended to help the scientific community, decision-makers, and sustainability analysts in understanding and utilizing the complex global energy data to foster climate action in line with the United Nations Sustainable Development Goal 13.

## II. PROBLEM STATEMENT

The growing global carbon dioxide (CO<sub>2</sub>) emissions is the main cause of climate change. This situation results mainly from increasing energy consumption, dependence on fossil fuels, and the uneven use of renewable energy technologies. Although the international community is working for decarbonization, the countries still show great differences in electricity access, energy efficiency, and use of renewable energy; as a result, there are huge variations in emissions all over the world. Global development indicators reveal that these disparities are closely linked to the social and economic characteristics, technological capacity, and national energy structures [2].

One of the challenges in solving the emission problem is understanding the complex nonlinear interactions of different energy, environmental, and economic factors. Traditional statistical methods cannot fully capture these multidimensional relationships. A recent empirical work suggests that machine learning techniques like Random Forest, Support Vector Regressor (SVR), Gradient Boosting Regressors, and Linear Regression not only offer better predictive performance but also can be used to pinpoint the most influential contributors to carbon emissions in different countries with high accuracy [1]. Nevertheless, these sophisticated

analytical instruments are still out of reach for many policymakers, students, and sustainability analysts because of technical difficulties and the absence of integrated visualization platforms.

Hence, the major issue that this research is tackling is the lack of a user-friendly, data-driven system that can predict CO<sub>2</sub> emission categories by employing several energy and economic indicators. Stakeholders, thus, are unable to make rational, data-driven decisions for sustainable energy planning and climate mitigation due to the absence of such a tool. This study takes a step to close the gap by creating a machine learning-based tool which can be used to classify and predict carbon emission levels from the international sustainable energy datasets [3] and, therefore, be a vehicle for the implementation of the climate policy in line with the United Nations Sustainable Development Goal 13 (Climate Action).

## III. RELATED WORKS

The study review a broad set of machine learning applications aimed at predicting carbon emissions at various scales, by looking at different algorithms and methods used in studies of both global and national emissions. Their review points out that traditional statistical models very often fail to capture complex and nonlinear relationships that are inherent in environmental, economic, and energy datasets. To overcome this shortcoming, the authors explore how machine learning methods - such as Random Forest, Support Vector Machines, Gradient Boosting Models, and Long Short-Term Memory (LSTM) networks - can be effective in dealing with so many variables that describe demographic growth, industrial production, renewable energy use, and urbanization degree. In fact, they show that on the one hand, ensemble and on the other hand, deep learning models are in general better than classical methods with the exception of small-scale datasets, in which the interaction of the features is less pronounced. [4]

Besides model performance, Zhang et al. also pinpoint the significance of feature selection and preprocessing from which facts lead to the conclusion that features such as fossil fuel consumption, electricity demand, and GDP frequently have the greatest impact on prediction results. Those issues they also bring up are data quality, time variations, and differences in national reporting methods. After facing these challenges they come to the decision that machine learning has become a key tool for emission forecasting and sustainability analytics due to its adaptability and accuracy. Hence, the research that you are carrying out with Sustainalyze using similar ML methods for the categorization and analysis of carbon emission intensity from multi-indicator datasets is closely connected to this. [4]

This study examine the role of artificial intelligence (AI), in particular, machine learning (ML), in predicting the carbon footprint of regions and sectors. Various ML algorithms such as Random Forest, Decision

Trees, and Artificial Neural Networks were used in their study to identify changes in environmental, socioeconomic, and energy-use from the data. The authors suggest that machine learning enables very significant improvements to traditional regression techniques, which can be particularly advantageous in identifying non-linear relationships between emissions and transport activity, industrial growth, electricity use, and household consumption patterns. As ML models achieve higher accuracy and stronger generalization capabilities, the authors argue that AI-based instruments can be extremely valuable in monitoring sustainability metrics and providing guidance for emission-reduction strategies. [5]

The paper also turns to the interpretability aspect of ML by examining feature importance in their models, besides the performance of ML models. The investigation revealed that the factors that led to the increase of CO<sub>2</sub> footprints were mainly fossil fuel consumption, energy consumption per capita, and economic output. The feature of interpretability becomes even more significant at this level to the policy makers and sustainability planners, who are seeking for the insights that they can act upon, not just the predictions. Sustainalyze is doubly important in this regard—your platform is equally reliant on identifying which energy and economic indicators exert the strongest influence on emission intensity classifications. Thus, Meng and Noman's findings align with your environmental analytics work being well-suited for ML and they back up your methodological decisions. [5]

Machine learning models combined with building energy data was used to solve carbon compliance forecasting and energy performance evaluation problems in commercial buildings. Their work involves applying state-of-the-art algorithms such as Support Vector Machines, Artificial Neural Networks, and a hybrid physics-based + ML strategy for emissions prediction at the building level and the compliance assessment with environmental regulations. They show that ML models beat traditional rule-based systems in most cases, particularly in the recognition of temporal patterns such as daily load variations, seasonal heating/cooling, and occupancy-driven energy use. Also, their hybrid modeling approach leads to a higher level of agreement which results from the utilization of the benefits of both domain-specific physical models and data-driven prediction. [6]

However, the authors, whose work is centered on building-level data rather than country-level emissions, discuss considerable methodological principles that are instrumental for large-scale sustainability analytics. Some of these principles involve the significance of data preprocessing pipelines, feature engineering, and model tuning which are the fundamental parts of emission prediction systems. Also, the integration of prediction with visual analytics tools, as emphasized by them, is very close to the aim of Sustainalyze, which is to provide a user-friendly, interactive platform for emission classification and analysis. Their research, in the end, conveys the message that ML-based forecasting systems

can be an essential source of support in the decision-making process of climate action, whatever the scale is, be it the building, city, or national level. [6]

This study examines different machine learning methods to interpret carbon emissions at the national level based on a multifactor dataset that includes energy, economic, and demographic indicators. Their work heavily revolves around Support Vector Machines (SVM), Gradient Boosting, and other sophisticated ML techniques that can detect nonlinear and high-dimensional interactions. The research shows that models based on SVM are particularly powerful in capturing complex national variation and, therefore, can be used for global comparative emissions analysis. [7]

Their results suggest that socioeconomic factors such as industrial energy consumption, transportation fuel usage, and economic development level have a strong impact on emission outputs. What is more, the authors emphasize that ML-based modeling should be considered as a tool for prediction and also as a method for identifying variable importance — an indispensable feature for policymakers who want to allocate intervention strategies efficiently. These revelations in turn confirm the core of your research, which is based on a multi-indicator approach for machine learning to classify countries into emission categories. [7]

This study is widely recognized as a key technique in revealing how different factors affect carbon dioxide emissions at various emission levels. Unravel the differential impacts of economic, energy, and environmental variables on carbon emissions. Their analysis showed that the influence of some variables varies across different emission levels. For instance, the effects of industrial activity, energy consumption, and renewable energy uptake on emission changes were significantly different in the lower, median, and upper quantiles [8]. Such an approach reveals how crucial it is to account for multiple levels of emissions rather than just the average ones.

By presenting evidence that effects differ across the distribution of emissions, researchers justify the use of percentile-based thresholds as a natural way to obtain Low, Medium, and High emission classes. Quantile division of emissions offers a more detailed picture of the factors that drive emissions at different levels, thus leading to a better understanding of the phenomena. The methodology used ensures that each class reflects a significant part of the emission distribution and that the models are more interpretable and actionable. Percentile thresholds like 33rd and 66th percentiles provide a coherent way of defining categories of emission intensity that is consistent with the heterogeneous effects found in their study [8].

For machine learning-based prediction models, using a percentile-based categorization of data is a step toward making predictions more operationally feasible. It is a quick means for decision-makers to identify areas or

countries with relatively low, medium, or high emissions and to have a clue of the factors that are causing such levels. The study by Khan et al. offers empirical support for the emission categorization process to be structured and data-driven which facilitates the interpretability of the predictive models in the field of climate action and sustainability planning [8].

#### IV. METHODOLOGY

This part of research utilizes the development of a machine learning-driven analytical pipeline for the classification and prediction of carbon emission intensity from global energy and economic variables. The analytical pipeline components consist of a process whereby real-time data sources are developed, the data is preprocessed, features are developed, a Reduced Row Echelon form is developed, the model is trained, verification is completed, and the analytically generated emissions data is deployed. Each component is made using Python machine-learning libraries on Google Colab. The construction of this methodological approach allows for reproducibility, scalability, and transparency. Therefore, the Sustainalyze system is an effective statistical model for estimating and adequately analyzing nonlinear relationships regarding global energy outcomes with newly generated estimation and emissions data, along with generating future emissions category predictive systems.

##### A) Data Preprocessing

The preprocessing phase prepares the raw international energy data for use in a clean, internally consistent, state and ready for machine learning analysis. The initial step is to load the dataset, confirm its structure, rename important variables, and delete columns that are not directly useful for prediction. For this study the Entity and Year columns were deleted to limit text noise in the country name and temporal granularity. Simple exploratory checks including the shape of the dataset, column data types, and counts of missing values were conducted to articulate what the data 'looked like' at the beginning.

To promote the validity of the learning process, the data went through several cleaning steps. Missing values, which are not uncommon when measuring global energy indicators, were filled with the median of each numeric column. This avoids disrupting the learning process from gaps in the records and avoids distortion due to extreme value effects. Duplicate rows were also checked and removed to avoid bias when training the model. Some of the features were housed in strings that included commas, and these were also changed into numeric values for consistency. To provide consistency in numeric variables, all numeric variables were standardized, allowing them to share a similar range of values in the model and prevent any one feature from dominating the model simply due to its magnitude.

```
# --- Libraries ---
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
import re

# --- Load Dataset ---
df = pd.read_csv("/content/global-data-on-sustainable-energy.csv")
print("Dataset Loaded. Shape:", df.shape)

# Show basic information about the dataset
print("\n★ DATASET INFO:")
df.info()

# Preview the first rows
df.head()

df = df.rename(columns={"Value_co2_emissions_kt_by_country": "Co2_emission"})
```

Figure 1. Importing Libraries Code

```
df = df.drop(columns=['Entity', 'Year'], errors='ignore')
print("\n✅ Dropped 'Entity' and 'Year' columns.")
print("Remaining columns:", df.columns.tolist())
```

Figure 2. Dropped Columns for Dataset

```
print("\nMissing values per column:")
print(df.isnull().sum())
missing_percent = (df.isnull().sum() / len(df)) * 100
plt.figure(figsize=(10,4))
missing_percent[missing_percent > 0].sort_values(ascending=False).plot(kind='bar')
plt.title("Missing Values (%) per Column")
plt.show()
```

Figure 3. Handling Missing Values Code

The dataset was analyzed for correlations of which features share strong positive/negative correlations with CO<sub>2</sub> emissions, to define the most relevant items and information for later model interpretation. The correlations were plotted in a bar graphed for visualization understanding. After the cleaning process had been completed, the finished data set was saved in a csv file labeled: ENERGY\_DATA\_CLEANED.csv, to be used for the model training and classification phase of the Sustainalyze system.

```
target="Co2_emission"

# Compute correlation
corr = df.corr(numeric_only=True)[target].sort_values(ascending=True)

print("Exact Correlation Values:\n")
print(corr)

# Colors: red for negative, green for positive
colors = ['red' if val < 0 else 'green' for val in corr]

plt.figure(figsize=(12, 8))
plt.barh(corr.index, corr.values, color=colors)
plt.title("Correlation with CO2 Emissions (Bar Graph)")
plt.xlabel("Correlation Value")
plt.ylabel("Features")
plt.grid(axis='x', linestyle='--', alpha=0.5)
plt.show()
```

Figure 4. Correlation Analysis Code

##### B) Loading The Dataset

After the dataset clean-up completed in the preprocessing step resulted in a cleaned dataset being loaded into the system in this modeling step, the focus no longer centers on the clean-up of the raw data, but the necessary preparation of the variables and for the machine learning modeling. The CO<sub>2</sub> emission values is the variable of interest or target variable, whereas the remaining indicators serve as the input variables or features of interest. Since some of the columns can still

be in string format formatted numbers for example, the coding incorporates one last conversion step to ensure all the features are treated properly as numeric. This is crucial to ensure the data is fully usable with the machine learning models being used for the subsequent stages with the study.

Once the dataset was verified to appropriately incorporate numeric values, the values were standardized through feature scaling. This process is important to provide uniformity of range; that is, the values of all indicators are similar in range to prevent any one variable from skewing results toward one variable over others. After the scaling of features, the dataset is then separated into a training and testing dataset, with 80% allocated to learning and 20% set aside for the model performance step. This separation provides fairness for assessing the accuracy of the model in predicting new/unknown data. At this point the features are fully prepared for the regression and classification modeling steps.

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# --- Load cleaned dataset ---
df = pd.read_csv("/content/ENERGY_DATA_CLEANED.csv")

# --- Features and target ---
X = df.drop(columns=["Co2_emission"]) # only drop CO2 target
y_reg = df["Co2_emission"]

# --- Convert any string numbers to numeric ---
for col in X.columns:
    if X[col].dtype == 'object':
        X[col] = X[col].str.replace(",", "", regex=False) # remove commas
        X[col] = pd.to_numeric(X[col], errors='coerce') # convert to numeric

# --- Fill missing values ---
X = X.fillna(X.median(numeric_only=True))

# --- Scale features ---
scaler = StandardScaler()
X_scaled = pd.DataFrame(scaler.fit_transform(X), columns=X.columns)

# --- Split data into train/test ---
X_train, X_test, y_train_reg, y_test_reg = train_test_split(
    X_scaled, y_reg, test_size=0.2, random_state=42
)

print("✅ Features converted, scaled, and split successfully.")
```

Figure 5. Dataset Loading Code

## C) Model Evaluation & Selection

### I. Regression Models

As a first step in selecting a suitable regression algorithm to predict CO<sub>2</sub> emission levels, this study explored four different modeling approaches: Linear Regression, Random Forest Regressor, Gradient Boosting Regressor, and Support Vector Regressor (SVR). Linear regression was selected as a baseline model because it is straightforward and can be easily interpreted, and, therefore, it acts as a starting point for assessing the results of the other more advanced algorithms. Random Forest and Gradient Boosting represent advanced ensemble methods capable of capturing complex nonlinear relationships; Random Forest proceeds by building many independent trees, while Gradient Boosting builds trees sequentially to correct the previous tree's errors. SVR was included as an approach to better capture more subtle and nonlinear patterns using learning based on margins.

Each modeling approach was trained with the scaled training dataset and then evaluated using the unseen data with two calculated metrics: Mean Squared Error (MSE) and R<sup>2</sup> score. The Mean Squared Error (MSE) quantifies the average difference between predicted and actual (verifiable) CO<sub>2</sub> values (MSE serves as an observed measure of prediction accuracy), whereas R<sup>2</sup> explains the amount of variance in CO<sub>2</sub> emissions that a model captures. By comparing metrics, both training and testing, these two observations ultimately allowed the researchers to evaluate overall accuracy and the likelihood of a model to overfit or underfit the data.

The review outcomes were held in a systematic dictionary for a simpler presentation of comparison. This technique gave the researchers the opportunity to see very clearly which models captured the dataset better than others and which algorithms required additional optimistic adjustments. The study with these baseline results was able to highlight the most promising candidate models, specifically Gradient Boosting, to go forward with hyperparameter tuning and clarify that the eventual model selection is both research supported and methodologically valid.

```
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
from sklearn.svm import SVR
from sklearn.metrics import mean_squared_error, r2_score
import pprint

# Initialize models
reg_models = {
    "Linear Regression": LinearRegression(),
    "Random Forest": RandomForestRegressor(n_estimators=100, random_state=42),
    "Gradient Boosting": GradientBoostingRegressor(n_estimators=100, random_state=42),
    "SVR": SVR()
}

# Train and evaluate
reg_results = {}
for name, model in reg_models.items():
    model.fit(X_train, y_train_reg)

# Predictions
y_train_pred = model.predict(X_train)
y_test_pred = model.predict(X_test)

# Store results
reg_results[name] = {
    "Train MSE": mean_squared_error(y_train_reg, y_train_pred),
    "Train R2": r2_score(y_train_reg, y_train_pred),
    "Test MSE": mean_squared_error(y_test_reg, y_test_pred),
    "Test R2": r2_score(y_test_reg, y_test_pred)
}

# Show results
print("📊 Regression Model Results (Train & Test):")
pprint.pprint(reg_results)
```

Figure 6. Regression Models Code

### II. Random Search For Gradient Boosting

To improve the predictive quality of the Gradient Boosting Regressor, this research included a process of hyperparameter tuning using Randomized Search. Gradient Boosting models are very sensitive to hyperparameter selection, including the number of trees, learning rate, maximum depth, minimum number of samples to split, leaf size, and subsample ratio of samples. Due to this sensitivity to hyperparameter selection, Randomized Search was used due to its efficiency in exploring hyperparameter combinations by randomly sampling values instead of exhaustively examining all combinations, which is an expensive process. Specifically, 20 random combinations were used with five-fold cross-validation, and the R<sup>2</sup> statistic was used as the scoring metric to ensure the model can generalize to other data splits.

Completing the random search, a detailed log was created from each parameter set with both training and testing performance. Each parameter set was retrained on the entire training datasets and tested on the reserved test set resulting in the calculation of metrics such as Mean Squared Error (MSE) and  $R^2$ . This logging process also provides transparency and reproducibility, allowing the researchers to see how different hyperparameters impacted their findings. All results recorded in a comprehensive Excel file to keep a complete record available for review and re-use in future experiments.

The hyperparameters that achieved the best performance through Randomized Search were applied to final fitting of the Gradient Boosting model. The final model was then assessed with the test dataset and, subsequently the MSE and  $R^2$  scores were reported confirming improvements from the baseline models. After carefully exploring the hyperparameter space the study confirmed that the selected model was accurate and robust, both factors which instill more confidence behind more accurate and robust CO<sub>2</sub> emissions prediction. This study shows how, when ensemble learning coupled with formalized hyperparameter tuning and optimization in predictive modeling can significantly improve predictive modeling in sustainability analytics.

### III. Optimized Model

After thorough investigation and hyperparameter tuning, the study selected Gradient Boosting Regressor with fine-tuned hyperparameters as the ultimate model, under conditions of 400 estimators, optimization learning rate of 0.05, maximum depth of 10 and subsample ratio of 0.90 to predict CO<sub>2</sub> emissions. The selected configuration works well as it allows for efficient computation and captures complex nonlinear relationships among energy, economic, and environmental indicators while minimizing prediction error and improving the model's capacity to generalize to unseen data. To enhance the interpretability of results for stakeholders, continuous CO<sub>2</sub> emission predictions were recorded into categorical levels of Low, Medium, and High. The categorical levels of emission intensity (or output level) were determined on the basis of 33rd and 66th percentiles from the training dataset, demonstrating a systematic approach to classifying emission intensity. In the second stage of continued classification, the categorical indicators provide a mechanism for decision-makers to quickly identify countries or areas with low, medium, or high carbon emissions intensity, which improves the practical applicability of the Sustainalyze system for sustainability analysis and policy forecasters.

The model's performance was assessed on the test set with Mean Squared Error (MSE) and  $R^2$  score used as a measuring performance metric, both of which showed that predicted values are highly accurate. We also implemented a function to allow for evaluating new input data by automatically calculating numeric CO<sub>2</sub> predictions along with their corresponding category. An example shows how the hypothetical country-year input was processed from beginning to end, providing a visible guide to the functionality of the system while ensuring that the optimized Gradient Boosting model can yield reliable insights that are interpretable for making sustainability-related business decisions in the real world.

```
import pandas as pd
import numpy as np
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.model_selection import RandomizedSearchCV
from sklearn.metrics import mean_squared_error, r2_score

# --- Hyperparameter distributions ---
param_dist = {
    'n_estimators': [100, 200, 300, 400],
    'learning_rate': [0.01, 0.02, 0.05, 0.1],
    'max_depth': [2, 3, 5, 7],
    'min_samples_split': [2, 5, 10],
    'min_samples_leaf': [1, 2, 4],
    'subsample': [0.5, 0.75, 0.9, 1.0]
}

# --- Initialize model ---
gbr = GradientBoostingRegressor(random_state=42)

# --- Random Search ---
random_search = RandomizedSearchCV(
    estimator=gbr,
    param_distributions=param_dist,
    n_iter=10,          # number of random trials
    scoring='r2',
    cv=5,
    random_state=42,
    n_jobs=-1,
    verbose=2
)

# --- Fit Random Search ---
random_search.fit(X_train, y_train_reg)

# --- Extract random search results ---
cv_results = pd.DataFrame(random_search.cv_results_)

# --- Build detailed logs ---
log_rows = []

for i, row in cv_results.iterrows():
    params = row['params']
    run_id = i + 1

    # Fit model with this parameter set
    model = GradientBoostingRegressor(random_state=42, **params)
    model.fit(X_train, y_train_reg)

    y_train_pred = model.predict(X_train)
    y_test_pred = model.predict(X_test)

    log_row = {
        'run_id': run_id,
        **params,
        'train_mse': mean_squared_error(y_train_reg, y_train_pred),
        'train_r2': r2_score(y_train_reg, y_train_pred),
        'test_mse': mean_squared_error(y_test_reg, y_test_pred),
        'test_r2': r2_score(y_test_reg, y_test_pred)
    }
    log_rows.append(log_row)

# --- Save logs to Excel ---
log_df = pd.DataFrame(log_rows)
excel_path = "/content/gradient_boosting_random_search_logs.xlsx"
log_df.to_excel(excel_path, index=False)

print(f"Random search logs saved to: {excel_path}")

# --- Best model evaluation ---
best_params = random_search.best_params_
best_model = random_search.best_estimator_

y_test_best = best_model.predict(X_test)
best_mse = mean_squared_error(y_test_reg, y_test_best)
best_r2 = r2_score(y_test_reg, y_test_best)

print("\n🌟 Best Hyperparameters:", best_params)
print("📊 Best Test MSE: (best_mse:.4f), Best Test R²: (best_r2:.4f)")
```

Figure 7. Random Search For Best Hyperparameter Code



```

from sklearn.ensemble import GradientBoostingRegressor
import numpy as np
import pandas as pd
from sklearn.metrics import mean_squared_error, r2_score

# --- Step 1: Train Gradient Boosting Regressor (UPDATED WITH RUN 2 PARAMS) ---
gb_reg = GradientBoostingRegressor(
    n_estimators=400,
    learning_rate=0.05,
    subsample=0.9,
    max_depth=10,
    min_samples_split=4,
    min_samples_leaf=2,
    random_state=42
)

gb_reg.fit(X_train, y_train_reg)

# --- Step 2: Define thresholds based on training set ---
low_thresh = np.percentile(y_train_reg, 33) # 33rd percentile + Low
high_thresh = np.percentile(y_train_reg, 66) # 66th percentile + High

def co2_category(pred):
    """Convert numeric CO2 emission to Low/Medium/High category."""
    if pred <= low_thresh:
        return "Low"
    elif pred <= high_thresh:
        return "Medium"
    else:
        return "High"

# --- Step 3: Predict on test set ---
y_pred_test = gb_reg.predict(X_test)
y_pred_category = [co2_category(val) for val in y_pred_test]

# --- Step 4: Evaluate numeric regression ---
mse = mean_squared_error(y_test_reg, y_pred_test)
r2 = r2_score(y_test_reg, y_pred_test)
print(f"Gradient Boosting Regression (Run 2 Params) + Test MSE: {mse:.2f}, R2: {r2:.4f}\n")

# --- Step 5: Show predicted CO2 with categories for first 10 test samples ---
print(f"Predicted CO2 + category (first 10 test samples):")
for val, cat in zip(y_pred_test[:10], y_pred_category[:10]):
    print(f"({val:.2f}) + {cat}")

```

Figure 8. Optimized Model Code Part 1

```

# --- Step 6: Function for new input prediction ---
def predict_co2_category(new_input_dict):
    """
    new_input_dict: dict with feature values, keys must match training columns (X_train)
    Returns: predicted CO2 emission and category
    """
    # Convert dict to DataFrame
    input_df = pd.DataFrame([new_input_dict])

    # Reorder columns to match training
    input_df = input_df[X_train.columns]

    # Scale features
    input_scaled = scaler.transform(input_df)

    # Predict CO2 emission
    pred_value = gb_reg.predict(input_scaled)[0]

    # Determine category
    pred_cat = co2_category(pred_value)

    return pred_value, pred_cat

# --- Step 7: Example usage ---
new_sample = {
    "Access to electricity (% of population)": 95.0,
    "Access to clean fuels for cooking": 75.0,
    "Renewable-electricity-generating-capacity-per-capita": 0.5,
    "Financial flows to developing countries (US $)": 2000000,
    "Renewable energy share in the total final energy consumption (%)": 30.0,
    "Electricity from fossil fuels (TWh)": 300.0,
    "Electricity from nuclear (TWh)": 50.0,
    "Electricity from renewables (TWh)": 100.0,
    "Low-carbon electricity (% electricity)": 40.0,
    "Primary energy consumption per capita (kWh/person)": 1500,
    "Energy intensity level of primary energy (kg/$2017 PPP GDP)": 6.0,
    "Renewables (% equivalent primary energy)": 20.0,
    "gdp_growth": 3.0,
    "gdp_per_capita": 40000,
    "Density (P/km2)": 300,
    "Land Area(km2)": 50000,
    "Latitude": 10.0,
    "Longitude": 120.0
}

pred_value, pred_cat = predict_co2_category(new_sample)
print(f"\nNew Input Prediction + CO2: (pred_value:.2f), Category: {pred_cat}")

```

Figure 9. Optimized Model Code Part 2

#### D) Saving Results

To enable the reusability of the trained model without needing to retrain, the final form of the Gradient Boosting Regressor was saved to disk using the Python pickle library. This process is efficient because it serializes the entire model, along with any learned parameters and its structure, into one file that can be loaded when performing predictions on new data in the future. Ultimately, saving the model creates a more reproducible output and enables facilitating the system in real-world implementations.

The feature scaler used to standardize the input variables during training was saved to disk. The Gradient Boosting model is designed to receive scaled inputs, and being able to save the scaler ensures that when new data are provided to the system, those data will be scaled in the same way as the training inputs so that correct predictions can be made. This function becomes particularly important when designing an operational model for real-world use, given that new country-year energy data may be consistently added over time and analyzed.

The cutoff points that were used to turn numeric CO<sub>2</sub> predictions into categories of Low, Medium, and High were also written to the output files. By saving the 33rd and 66th percentiles, the model guarantees that the categorization will always remain consistent for future input. Saving the model, scaler, and cutoff points provides a complete package to achieve reproducible, interpretable and reliable predictions that will help enable stakeholders to effectively employ the Sustainalyze system for sustainability and policy planning.

```

import pickle

# --- Save the Gradient Boosting model ---
with open("gb_reg_model.pkl", "wb") as f:
    pickle.dump(gb_reg, f)

# --- Save the scaler ---
with open("scaler.pkl", "wb") as f:
    pickle.dump(scaler, f)

# --- Optional: Save thresholds too (for CO2 categories) ---
thresholds = {"low": low_thresh, "high": high_thresh}
with open("co2_thresholds.pkl", "wb") as f:
    pickle.dump(thresholds, f)

print("✅ Model, scaler, and thresholds saved as .pkl files!")

```

Figure 10. Saving Results Code

## V. RESULTS

### a) Random Search Training Method For Gradient Boosting

| Run_ID | subsample | n_estimators | min_samples_split | min_samples_leaf | max_depth | learning_rate | Train_MSE   | Train_R2    | Test_MSE    | Test_R2     |
|--------|-----------|--------------|-------------------|------------------|-----------|---------------|-------------|-------------|-------------|-------------|
| 1      | 0.6       | 400          | 10                | 4                | 3         | 0.01          | 9847691703  | 0.983353885 | 3142309147  | 0.988974816 |
| 2      | 0.9       | 400          | 10                | 4                | 2         | 0.05          | 2244373448  | 0.996206208 | 1970050373  | 0.993087833 |
| 3      | 0.9       | 300          | 5                 | 4                | 7         | 0.01          | 4233597172  | 0.992843709 | 7236426736  | 0.974610094 |
| 4      | 1         | 200          | 10                | 2                | 7         | 0.01          | 1166693237  | 0.981123896 | 14279040935 | 0.949902004 |
| 5      | 0.9       | 200          | 2                 | 2                | 7         | 0.05          | 20316775.69 | 0.999965657 | 3718256897  | 0.986954032 |
| 6      | 0.75      | 300          | 2                 | 1                | 7         | 0.1           | 295266.8049 | 0.999995501 | 498006348.1 | 0.998252683 |
| 7      | 0.9       | 100          | 10                | 1                | 7         | 0.001         | 4.85418E+11 | 0.179470758 | 2.37376E+11 | 0.167136535 |
| 8      | 0.9       | 200          | 10                | 2                | 2         | 0.05          | 2762521184  | 0.995330553 | 2230823456  | 0.992172878 |
| 9      | 0.9       | 300          | 10                | 1                | 5         | 0.001         | 3.27409E+11 | 0.446480506 | 1.6307E+11  | 0.425734426 |
| 10     | 0.9       | 300          | 10                | 1                | 5         | 0.05          | 31270630.46 | 0.999947141 | 2387930399  | 0.991621648 |
| 11     | 1         | 200          | 5                 | 4                | 5         | 0.1           | 200697447.8 | 0.99966075  | 6642512257  | 0.976693917 |
| 12     | 0.9       | 100          | 10                | 4                | 2         | 0.001         | 4.94746E+11 | 0.16370205  | 2.40568E+11 | 0.15593873  |
| 13     | 0.75      | 200          | 2                 | 4                | 7         | 0.05          | 858371274.7 | 0.998540406 | 1421499392  | 0.995012402 |
| 14     | 0.75      | 100          | 2                 | 2                | 3         | 0.01          | 89405574892 | 0.848872656 | 47294631752 | 0.834060885 |
| 15     | 1         | 300          | 5                 | 2                | 5         | 0.05          | 25655340.53 | 0.999956633 | 7112235858  | 0.975045833 |
| 16     | 1         | 400          | 2                 | 2                | 2         | 0.05          | 1275828340  | 0.997843395 | 1778275617  | 0.993760698 |
| 17     | 1         | 300          | 10                | 2                | 5         | 0.01          | 2050012747  | 0.996533225 | 721978307   | 0.974660788 |
| 18     | 0.6       | 100          | 10                | 1                | 2         | 0.05          | 7079758726  | 0.986511550 | 3720488758  | 0.986946198 |
| 19     | 1         | 300          | 5                 | 1                | 7         | 0.001         | 3.25464E+11 | 0.449850397 | 1.67588E+11 | 0.411997955 |
| 20     | 0.75      | 200          | 5                 | 1                | 2         | 0.001         | 4.1457E+11  | 0.299229456 | 1.96545E+11 | 0.310398896 |

Figure 11. Experimentation Logs

Random Search tried several different Gradient Boosting hyperparameter combinations to figure out how learning rates, tree depths, sample sizes, and ensemble sizes influence the predictive accuracy. The results varied significantly in twenty different runs which is an indication that the model is very sensitive to the parameter tuning. Moderate learning rates and deeper trees could reach higher R<sup>2</sup> values and lower MSE more often, while very small learning rates or shallow depths constrained the model's capacity to uncover the underlying patterns.

Most of the good configurations were able to generalize well as evidenced by low Train and Test MSE values as well as  $R^2$  values above 0.98. The model was in Run ID 6, which yielded a Test  $R^2$  of 0.99825 with a subsample of 0.75, 300 estimators, depth of 7, a 0.1 learning rate, and minimal split requirements. This combination of parameters successfully controlled model complexity and stability, therefore, it was possible to make very accurate predictions and thus, it can be considered as a perfect instance of well-tuned Gradient Boosting models operating efficiently.

### b) Gradient Boosting Model

```
*** Gradient Boosting Regression (Run 2 Params) -> Test MSE: 4434922137.38, R2: 0.9844

Predicted CO2 -> Category (first 10 test samples):
4612.16 -> Low
13259.89 -> Medium
12673.12 -> Medium
2869.02 -> Low
292388.45 -> High
154863.50 -> High
89638.43 -> High
178653.74 -> High
2328.27 -> Low
9662.13 -> Medium

New Input Prediction -> CO2: 448462.45, Category: High
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning:
warnings.warn(
```

Figure 12. Gradient Boosting Model Results

The Gradient Boosting Model turned out to be the final best predictor with the parameters as follows: 400 estimators, 0.05 learning rate, 0.9 subsample ratio, maximum depth of 10, and standard minimum-sample constraints. With this setup, the model was able to capture complex nonlinear relationships while the training remained stable. Hence, the model was very powerful, which is also indicated by the Test MSE value being close to 4.43 billion and the Test  $R^2$  equal to 0.9844, a signature of the model's high explanatory power of CO<sub>2</sub> emission variability.

In order to simplify the model, the continuous predictions were converted into three classes of CO<sub>2</sub> emissions that were derived from the 33rd and 66th percentiles of the training distribution. The predictions on the test set corresponded closely to these cutoffs, thus there was a distinct separation between the observations with low emissions and the outliers with high emissions. The categorical variables generated could be seen as the level of interpretation most relevant to the policymakers of emission differences.

The model was able to generalize to new data as well. The emission level of 448462.45 was predicted for a new example that was not used during training, which means that it belongs to the high-emission category and, therefore, the model is exhibiting consistent behavior even in the face of new conditions. To sum up, the Gradient Boosting Model is a powerful tool that combines accuracy, interpretability, and generalization to tackle large-scale environmental and energy prediction problems.

### c) Energy and Economic Indicators

Figure 13. Energy and Economic Indicators First Set

The Indicators revealed by the inputted set of energy and economic indicators shows that the access to electricity is at a normal level, access to clean cooking is extremely limited, and the capacity for renewables per capita is very low. The energy supply is described as having a very low nuclear contribution, almost no renewable electricity generation, and only minor shares of renewable and low-carbon sources in the total mix. Besides that, economic conditions like GDP per capita, population density, and land area also signal the expected volume of national energy use, which in turn, shows a system with limited economic output and moderate energy demand rather than a heavy industrial one. The combined inputs serve as a structured framework to quantify the country's potential CO<sub>2</sub> emissions under normal consumption and production patterns.

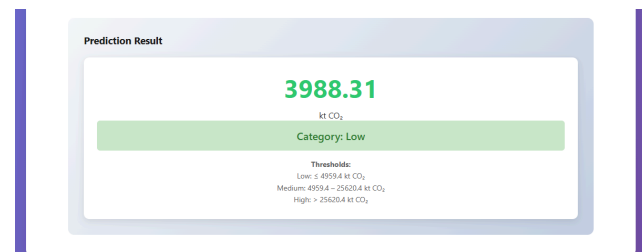


Figure 14. Energy and Economic Indicators Results Low

Under such circumstances, the model comes up with a forecasted emission volume of 3988.31 kt CO<sub>2</sub>, which is categorized as Low based on the set thresholds. This figure is still under the limit of 4,959.4 kt CO<sub>2</sub>, which indicates the extent of the lower emission range, thus showing that the energy configuration of the country is associated with less carbon output compared to the systems of higher consumption or fossil-dependency. The classification shows the limitations of how a small use of fossil fuel for electricity, moderate energy intensity, and a relatively small economic footprint can efficiently lead to lower emissions. The classification provides a straightforward understanding of the prediction, underlining that the given indicators correspond to a low-emission scenario within the model's learned distribution.



**Enter Energy & Economic Indicators**

|                                                                                       |                                                             |                                                                    |
|---------------------------------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------|
| Access to electricity (% of population)<br>70 %                                       | Access to clean fuels for cooking (% of population)<br>30 % | Renewable capacity per capita (kilowatt per person)<br>1 kW/person |
| Financial flows (US Dollars (USD))<br>500000 USD                                      | Renewable energy share (percent (%))<br>20 %                | Fossil fuel electricity (TeraWatt-Hours)<br>3 TWh                  |
| Nuclear electricity (TeraWatt-Hours)<br>0 TWh                                         | Renewable electricity (TeraWatt-Hours)<br>.5 TWh            | Low-carbon electricity (% of total electricity)<br>50 %            |
| Primary energy consumption (kilowatt-hour per person (kWh/person))<br>3000 kWh/person | Energy intensity (MegaJoules per Dollar)<br>15 MJ/\$        | Renewables share (percent (%))<br>20 %                             |
| GDP growth (percent (%))<br>3 %                                                       | GDP per capita (US Dollars (USD))<br>8000 USD               | Population density (persons per square kilometer)<br>120 P/Km²     |
| Land area<br>50000 Km²                                                                | Latitude (degrees)<br>15 °                                  | Longitude (degrees)<br>75 °                                        |

**Predict CO<sub>2</sub> Emissions**

Figure 15. Energy and Economic Indicators Second Set

The energy and economic attribute differences displayed in the second table indicate a country with a somewhat more robust developmental background than the first scenario, which can be seen from higher electrification, a more considerable share of primary energy consumption, and a larger GDP. The lack of clean cooking fuels is still a most apparent problem in the country; however, the small renewable electricity supply and the 50% low carbon share indicate some energy mix diversification. At the same time, fossil fuel electricity generation is still there, and higher energy intensity means that the economy is more energy-hungry per unit of output, thus is responsible for additional carbon emissions in general.

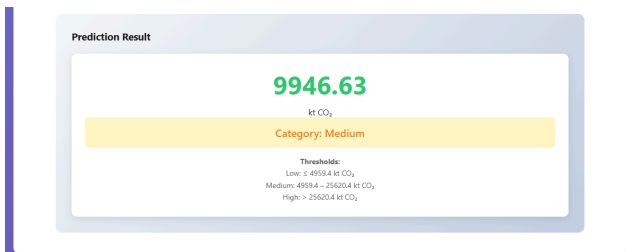


Figure 16. Energy and Economic Indicators Results Medium

Given these inputs, the model forecasts 9,946.63 kt CO<sub>2</sub>, ranking the country in the medium-emission group. The figure is more than double the low-emission threshold of 4,959.4 kt CO<sub>2</sub> but still far from the 25,620.4 kt CO<sub>2</sub> limit that separates high- from medium-emission groups. The mix of increased energy consumption per capita, moderate economic activity, and a partially fossil-dependent electricity supply is what drives emissions at the mid-range level. The classification mirrors the extent to which growing energy demand and economic expansion raise CO<sub>2</sub> emissions, even if some low-carbon and renewable elements are there, thus locating the country at the moderate-emission profile level in the model's learned distribution.

**Enter Energy & Economic Indicators**

|                                                                                       |                                                            |                                                                     |
|---------------------------------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------|
| Access to electricity (% of population)<br>95 %                                       | Access to clean fuels for cooking (% of population)<br>5 % | Renewable capacity per capita (kilowatt per person)<br>.5 kW/person |
| Financial flows (US Dollars (USD))<br>2000000 USD                                     | Renewable energy share (percent (%))<br>30 %               | Fossil fuel electricity (TeraWatt-Hours)<br>300 TWh                 |
| Nuclear electricity (TeraWatt-Hours)<br>50 TWh                                        | Renewable electricity (TeraWatt-Hours)<br>100 TWh          | Low-carbon electricity (% of total electricity)<br>40 %             |
| Primary energy consumption (kilowatt-hour per person (kWh/person))<br>1500 kWh/person | Energy intensity (MegaJoules per Dollar)<br>6 MJ/\$        | Renewables share (percent (%))<br>20 %                              |
| GDP growth (percent (%))<br>3 %                                                       | GDP per capita (US Dollars (USD))<br>40000 USD             | Population density (persons per square kilometer)<br>300 P/Km²      |
| Land area<br>50000 Km²                                                                | Latitude (degrees)<br>10 °                                 | Longitude (degrees)<br>120 °                                        |

**Predict CO<sub>2</sub> Emissions**

Figure 17. Energy and Economic Indicators Third Set

The third set of indicators reflects a country with electrification rate of around 75 percent and access to clean cooking fuels is very limited, only 5 percent. The energy profile contains 300 TWh of electricity from fossil fuel sources that are complemented by 50 TWh from nuclear and 100 TWh from renewable sources. Hence, the share of low carbon electricity is 40 percent. The country's primary energy consumption is 1,500 kWh per capita and the energy intensity level of 6 MJ per dollar points to a relatively energy-consuming economy. The socioeconomic indicators such as GDP per capita of 40,000 USD, GDP growth of 3 percent, and moderate population density describe a system with high energy consumption and continuous carbon-intensive source reliance.

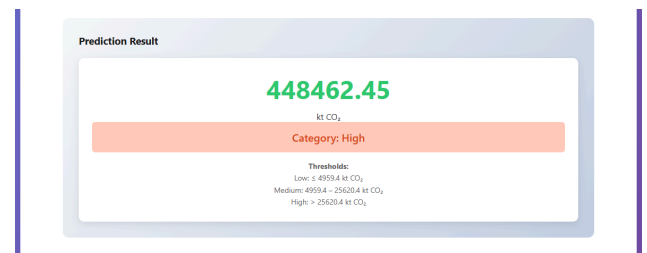


Figure 18. Energy and Economic Indicators Results High

With these inputs, the model estimates emissions at 448,462.45 kt CO<sub>2</sub>, which categorizes the country as a high emitter. The value is way above the upper limit of 25,620.4 kt CO<sub>2</sub>, indicating a substantial carbon footprint that is mostly driven by high electricity production and economic activity. Although the country has partially diversified its energy mix by using nuclear and renewables, it seems like the main reasons for the high emission level are the overall energy demand and the continued use of fossil fuels.

## VI. DISCUSSION

The study aimed to investigate how multiple sustainable-energy indicators influence CO<sub>2</sub> emissions and to identify the most effective machine-learning model for accurate prediction and classification. The dataset underwent extensive preprocessing, including handling missing values using median imputation, removing duplicates, converting non-numeric columns, and

applying feature scaling. These steps ensured that the data was clean, consistent, and suitable for machine-learning modeling. A correlation analysis revealed that features such as electricity from fossil fuels, primary energy consumption, and gdp\_per\_capita had strong positive associations with CO<sub>2</sub> emissions, confirming established energy-economic trends.

Four regression models were evaluated, Linear Regression, Random Forest, Gradient Boosting, and Support Vector Regression (SVR). The performance evaluation showed that Gradient Boosting and Random Forest significantly outperformed the other models, achieving R<sup>2</sup> scores above 0.99, while Linear Regression recorded moderate performance, and SVR failed to model the relationships (R<sup>2</sup> < 0). Gradient Boosting emerged as the optimal model due to its excellent balance of generalization ability and predictive accuracy, achieving Train R<sup>2</sup> = 0.9994 and Test R<sup>2</sup> = 0.9924 using default parameters. These results indicate that ensemble-based boosting effectively captured the nonlinear interactions between multiple energy features and CO<sub>2</sub> emissions.

To further enhance performance, Randomized Search CV was applied to tune key hyperparameters of Gradient Boosting. The Run 2 configuration demonstrated the most stable and accurate results, producing Test R<sup>2</sup> = 0.9844 with the tuned model. Although the tuned model's MSE increased slightly due to the more complex parameter set (max\_depth=10), the performance remained strong and highly reliable. The predictions on unseen data showed that the model was capable not only of predicting emission values but also of categorizing outputs into Low, Medium, and High emission groups using percentile-based thresholds. This categorization approach makes the system practical for real-world policy applications, where qualitative classification aids decision-making.

Finally, the selected Gradient Boosting model, scaler, and CO<sub>2</sub> classification thresholds were saved as .pkl files, demonstrating the system's readiness for deployment. The workflow, from data cleaning to model saving, ensures reproducibility and practical usability for future forecasting tasks.

## VII. CONCLUSION

This research successfully developed a machine-learning model capable of predicting and categorizing CO<sub>2</sub> emissions based on sustainable-energy indicators. Through systematic preprocessing, multi-model evaluation, and hyperparameter optimization, the study identified Gradient Boosting Regression as the most effective model. It achieved high predictive performance, with the best generalized model recording an R<sup>2</sup> of 0.9844 on unseen data.

The results demonstrate that ensemble-boosting methods are highly suited for modeling CO<sub>2</sub> emissions, where relationships are nonlinear and influenced by multiple interacting economic and energy factors. The

model's ability to classify emissions into Low, Medium, and High categories also enhances its usefulness for climate monitoring, sustainability planning, and SDG-related decision-support systems.

Future work may include integrating time-series forecasting, adding more structural economic indicators, or deploying the model in an interactive dashboard for continuous monitoring. Overall, the study provides a strong data-driven foundation for analyzing energy sustainability and its impact on global emissions.

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